EDUPORTAL AND EDUOS

Status: Pipeline

TARGET CAPITAL	AMOUNT SECURED	REMAINING REQUIRED
Rs.1,00,00,000	Rs.0	Rs.1,00,00,000

1. Project Overview

EduPortal and EduOS form a hybrid, offlin
for low-connectivity and rural school enviro
breaks when internet connectivity drops, c

By deploying a localized hardware-softwar
daily workflows—from registration to final e
Operating as "one calm control room for m
administrative chaos while acting as a com
2020) mandates.

2. Strategic Motive

EduPortal and EduOS unifies entire schoo
into a single, highly resilient application. Th
acting as a compliant vehicle for India's Na
connectivity rural school environments.

3. Detailed Specifications

1. EXECUTIVE SUMMARY

EduPortal and EduOS form a hybrid, offline-first edge computing architecture engineered specifically for low-connectivity and rural school environments. Traditional educational software completely breaks when internet connectivity drops, creating massive administrative bottlenecks. By deploying a localized hardware-software ecosystem on campus, this project unifies entire school daily workflows—from registration to final examinations—into a single, highly resilient application. Operating as "one calm control room for modern schools," the platform addresses structural administrative chaos while acting as a compliant vehicle for India's National Education Policy (NEP 2020) mandates.

2. PROBLEM STATEMENT & MARKET OPPORTUNITY

• The Connectivity Paradox: Schools across regional and Tier-2/3 sectors heavily desire modern, AI-powered digital tools but lack the persistent, high-bandwidth internet connectivity required to run traditional cloud software.

• Cloud Dependency Fragility: Pure-cloud platforms suffer total operational failure during internet blackouts, leaving teachers unable to access basic daily tools like attendance tracking or digital lesson plans.

• Administrative Fragmentation: School operations are structurally chaotic, forcing educators to juggle multiple disconnected applications for attendance, compliance, gradebooks, and student reporting.

• The Maintenance Nightmare: Deploying localized servers or complex IT networks in remote setups traditionally requires an extensive, highly expensive on-the-ground IT support presence.

• Capital Intensity: Hardware distribution models for educational technology are notorious for exhausting upfront capital without providing long-term operational scaling mechanisms.

3. PRODUCT & TECHNICAL ARCHITECTURE SPECIFICATION

The system replaces standard cloud-only models with a two-tiered, offline-resilient edge infrastructure.

• Class Station (Edge Server Node): Operates on the campus perimeter, hosting local services on http://127.0.0.1:4102/school/teacher. It connects to external networks only to execute cloud syncs or external cloud API calls, running flawlessly offline during dropouts by queuing database updates locally.

• Resilient Data Storage: Backed by 1.5TB of local storage. It features an automated First-In-First-Out (FIFO) cleaning mechanism to purge the oldest cached assets and prevent storage overflow. To protect against physical drive corruption, it enforces a minimum RAID 1 (Mirroring) disk setup.

• Student Hub (Intranet Client): Accessible locally on http://127.0.0.1:4101/school/student. It functions entirely within the school's local intranet perimeter with zero outbound public internet access required, serving as the interactive thin-client for student desks.

4. CORE WORKFLOWS & FUNCTIONAL MODULES

• Asynchronous "Local-to-Cloud" AI Workflow: A student or teacher submits a prompt (e.g., creating flashcards or summarizations) from an offline Student Hub thin-client. The Next.js API captures the request and inserts a pending row into the local Supabase database. The moment the Class Station catches an internet signal, a background worker handles the payload, securely executing remote cloud API endpoints (Gemini). The generated assets are downloaded, cached at the edge, and pushed to the Student Hub interface using real-time browser subscriptions.

• Interactive Media & AI Studio: The AI Studio module parses raw JSON schemas generated from notes and renders interactive slides or presentation decks completely inside the browser viewport, avoiding capital-heavy generation of physical .pptx files. Media transfers adapt to weak cellular connections by leveraging chunked, resumable file uploads.

• The Localized Exam Lifecycle: Live examinations are deployed and completed locally across Student Hubs without pinging public clouds. When connection permits, encrypted performance arrays push to the central cloud repository, triggering automated AI-driven OCR document grading.

5. UNIQUE SELLING PROPOSITION (USP) & POLICY ALIGNMENT

• The "Calm" Operational Control Room: The primary interface unifies staff management, lesson pacing, student registration, scheduling, and standard administrative overhead into one central dashboard. This gives administrators emotional relief from traditional, chaotic school management overhead.

• NEP-2020 Compliance Engine: "NEP-Ready Operations" act as an aggressive compliance wedge. 360-Degree Holistic Progress Cards aggregate academic testing, behavioral metrics, and continuous extracurricular development parameters. Competency-Based Assessment replaces traditional high-stakes testing matrices with formative metric tracking arrays evaluating critical thinking.

6. BUSINESS MODEL & GO-TO-MARKET STRATEGY

• Revenue Model: Sustainable B2B model tailored for educational groups and regional systems. Recurrent subscription modeling charged per student node for platform access, AI features, and compliance logging. Hardware Licensing Packages offer one-time or leased licensing models for localized edge-server equipment.

• Scalability via Fleet Management: Remote telemetry tracking layout lets central administrators trace the systemic health, compute load, disk capacity, and background sync logs of thousands of scattered edge systems from a remote cloud dashboard, ensuring hyper-scalability.

7. RISK MANAGEMENT & ENGINEERING Q&A

• Q: How does the system resolve data collisions if record modifications occur concurrently online and offline?

• A: When an offline Class Station triggers its data re-sync, entries are merged using Conflict-free Replicated Data Types (CRDTs) or explicit, timestamped vector clocks embedded in the Supabase schema layer.

• Q: What prevents student lockouts from localized interfaces during prolonged internet outages?

• A: EduOS implements a secure local auth-proxy on the Class Station that issues short-lived offline authentication tokens, keeping students verified on the local intranet indefinitely.

8. CURRENT IMPLEMENTATION STATUS & FINANCIAL ASK

• Repository Benchmarks: Active private repository tracking over 101 developmental commits, with successfully compiled deployment-ready image footprints for student-hub-v1.0.0.img and class-station-v1.0.0.img.

• Fund Requirement: Evaluating a 1 Crore (1CR) Convertible Note framework. Capital will fund hardware deployment provisioning, local quantized LLM integration fallbacks, and regional compliance expansion campaigns.